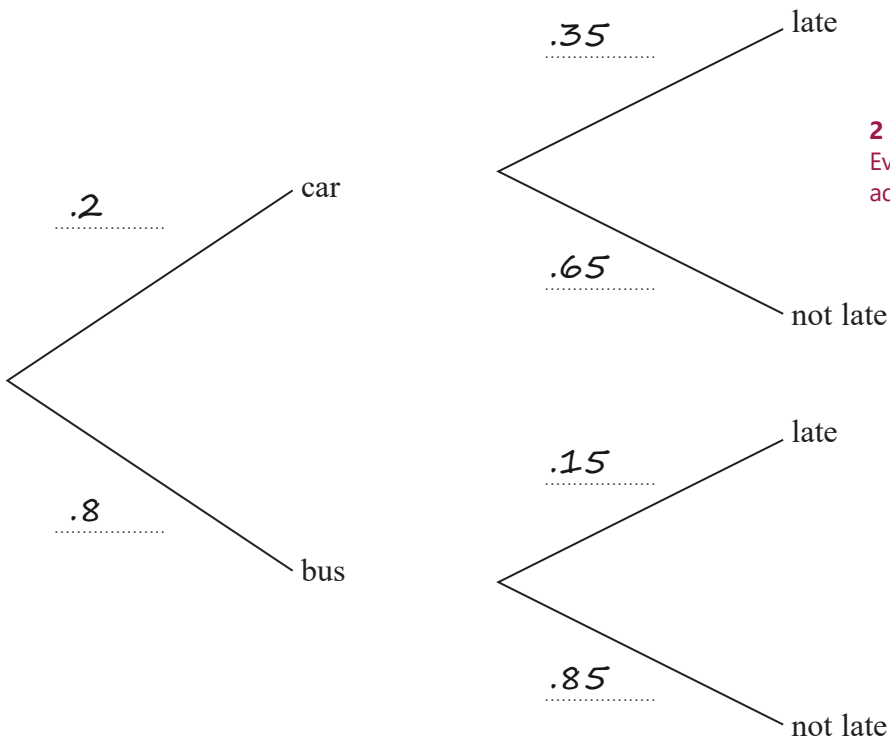
13 When Lee goes to work, he either travels by car or by bus.

The probability that on any day Lee goes to work he travels by car is 0.2

When Lee travels to work by car, the probability that he is late is 0.35
When Lee travels to work by bus, the probability that he is late is 0.15

(a) Complete the probability tree diagram for this information.



2 marks total
Every pair adds to1.

(2)

(b) Work out the probability that on any day Lee goes to work he is late.

Car and late: $0.2 \times 0.35 = 0.07$

1 mark
Multiply along a late route.

Bus and late: $0.8 \times 0.15 = 0.12$

1 mark
Add the two late routes.

$P(\text{late}) = 0.07 + 0.12 = 0.19$

1 mark
0.19.

(3)

(Total for Question 13 is 5 marks)